

1. Introduction

The ECP205 is a single DOF system torsional mechanical system. It has a DC motor and an encoder which measure the disk angle and motor input voltage. The friction at the bearings and shaft create damping.

The aim is to measure the transient response of the open loop and identify the parameters of the torsional system.

And analyse how feedback gain can change transient behaviour.

The objectives are to identify system parameters.

Analyse the effect of proportional k_p gain and effect of k_d gain for PD control.

Compare experimental data with MATLAB simulations.

2.1DOF Parameters

Rotational equation of motion:

$$J \cdot \alpha = T_A - c \cdot \omega - T_f$$

From experiment:

$$T_A = 0.6$$

$$J = T_A / (\alpha_1 - \alpha_3)$$

$$c = -J (\alpha_2 - \alpha_3) / \omega_2$$

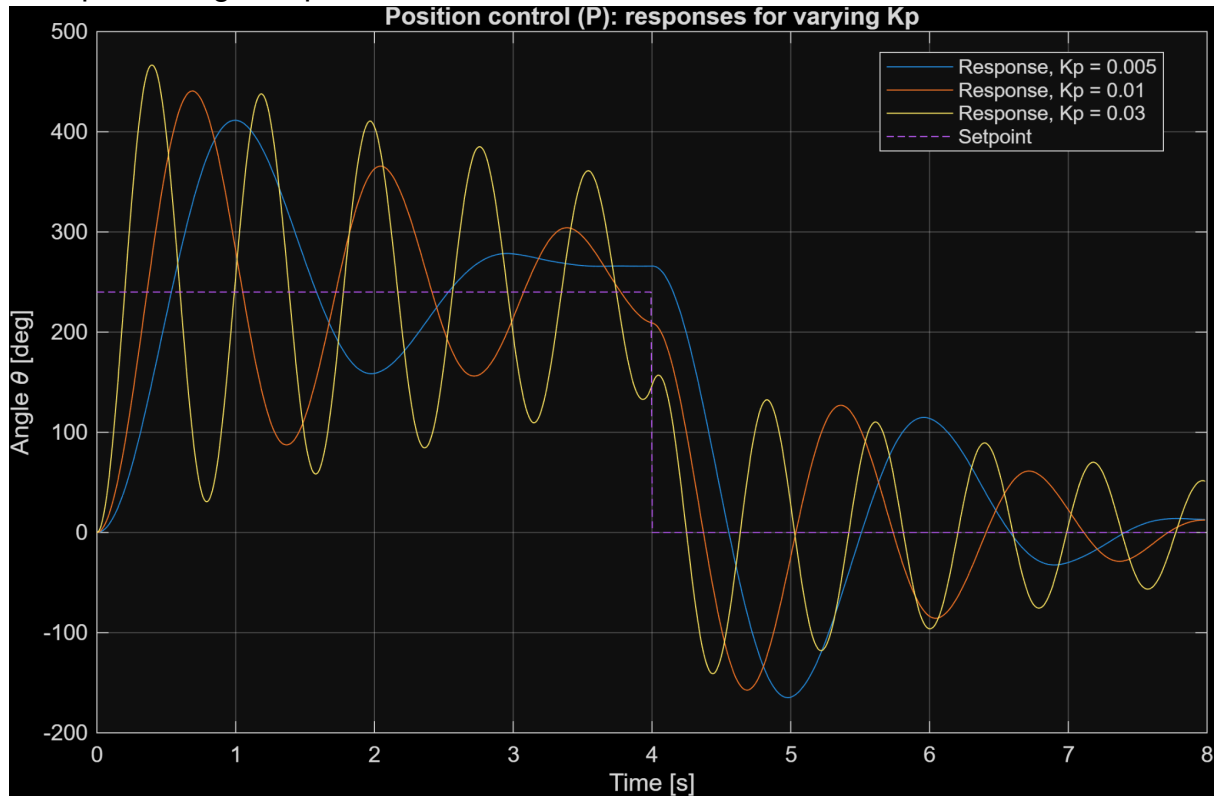
$$T_f = -J \alpha_3$$

$$J = 0.006972 \text{ kgm}^2$$

$$c = 0.002989 \text{ Nms}$$

$$T_f = 0.0357 \text{ Nm}$$

3. Proportional gain K_p



When the k_p increases the rise time decreases because the closed loop natural frequency increases. The overshoot increases and oscillations become frequent due to a reduction in damping ratio. When k_p is large values it increases instability.

4.

Kp	Overshoot_ pc t	PeakTime_ s s	RiseTime_ s s	SettlingTime_ s s	SSE_ deg g
0.005	92.4	4.985	0.236	7.561	-13.26
0.010	145.7	4.684	0.109		-4.06
0.030	537.8	4.436	0.019		-5.07

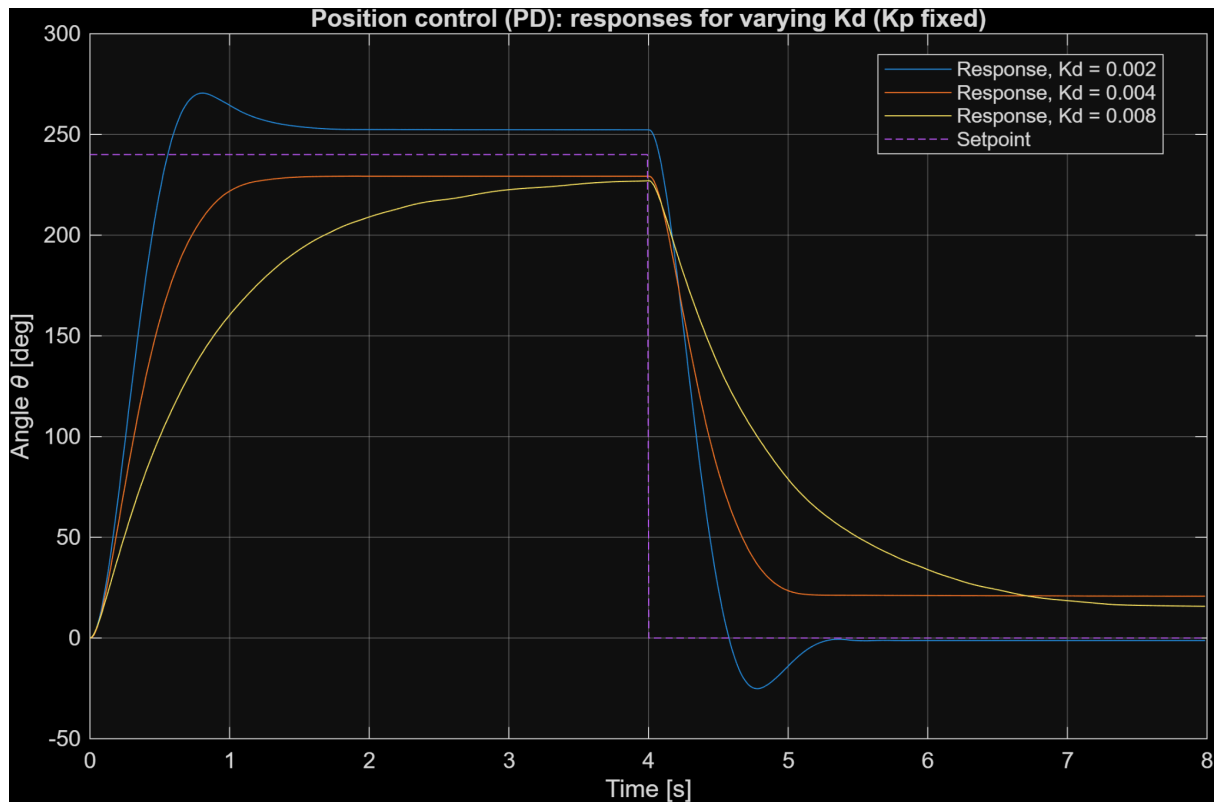
Increasing k_p causes:

Rise time decreases because the natural frequency in the closed loop increases

Overshoot increases because the damping ratio decreases and causes more bigger oscillations.

Instability risk increases.

5.



When k_d increases damping also increases which reduces oscillations and overshoot. When k_d becomes larger the rise time increases and can slow system response.

6.

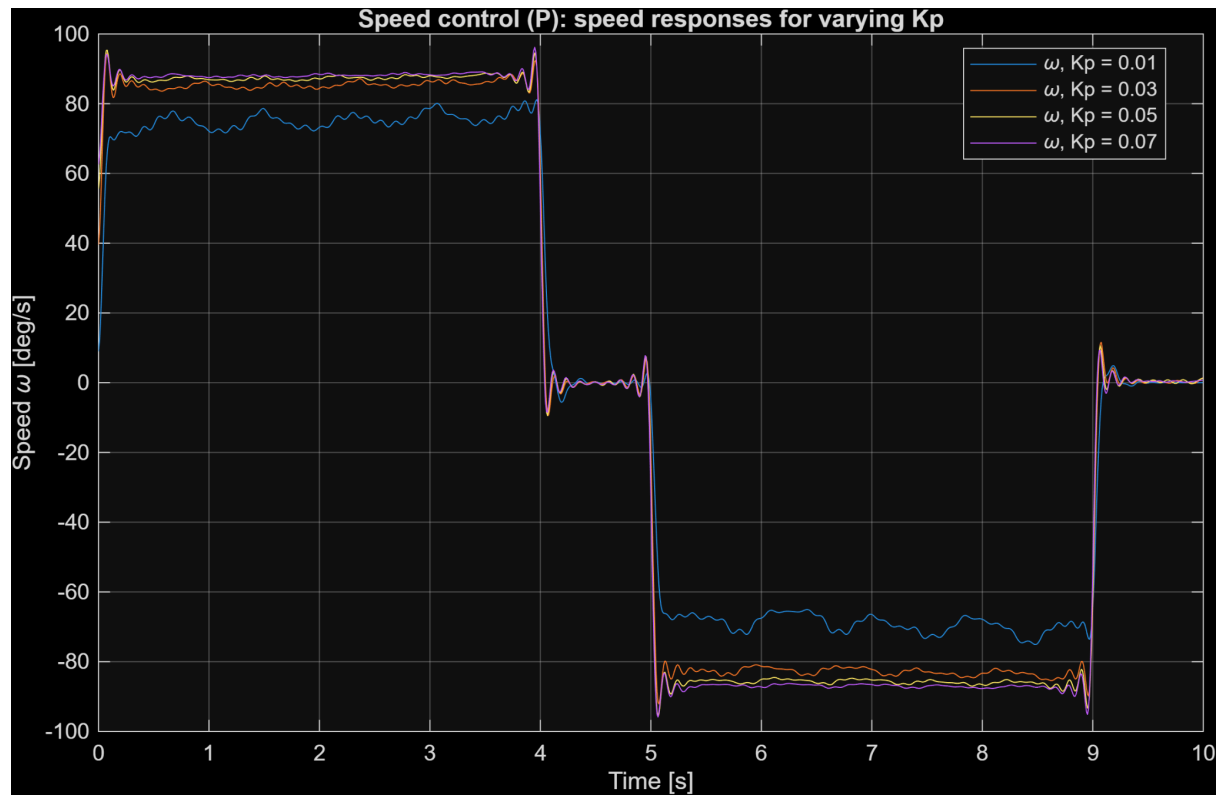
Kd	Overshoot_ pc t	PeakTime_ s s	RiseTime_ s s	SettlingTime_ s s	SSE_ deg g
0.00 2	13.82	4.781	0.274	5.162	1.25
0.00 4	0.01	7.986	0.528	4.994	-20.77
0.00 8	0.08	7.986	1.732	6.871	-15.91

Increasing kd causes:

Overshoot decreases because of effective damping

Rise time increases due to increased damping

7.



When k_p is increased the steady state speed error is reduced. However higher k_p also increases overshoot and sensitivity.

8.

Kp	Overshoot_pc t	PeakTime_ s	RiseTime_ s	SettlingTime_ s	SSE_deg_per _s
0.0 1	0.0	3.055	0.673	N/A	13.26
0.0 3	2.9	3.789	0.026	N/A	3.47
0.0 5	14.6	0.026	0.018	N/A	2.19
0.0 7	17.9	0.018	0.009	N/A	1.54

Rise time decreases with increasing k_p .

Steadystate error decreases.

Overshoot and oscillations increase when k_p is large.

9.

Plant model:

$$G(s) = 1 / (J s^2 + c s)$$

Controller:

$$C(s) = K_p + K_d s$$

Close loop transfer function:

$$T(s) = C(s)G(s) / [1 + C(s)G(s)]$$

$$T(s) = (K_p + K_d s) / (J s^2 + c s + K_d s + K_p)$$

Substituting:

$$T(s) = (K_p + K_d s) / (0.006972 s^2 + 0.002989 s + K_d s + K_p)$$

10.

The torsional plant behaves as a damped rotational system. The system parameters were identified using experimental data. When k_p is increased it improves response speed but increases overshoot and oscillations. The steady state error is reduced but having large values increases sensitivity.

Recommendations:

Repeat the experiment again and average results by calculating mean.

Use higher resolution encoders.